

Downside Skew and HAR Forecast Errors

Evidence from a pseudo-out-of-sample SPX exercise

VRP-Skew Analysis Project

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Abstract

This project began with a pattern in the SPX data: HAR forecast errors were much more variable when downside option-implied skew was high. I therefore added skew to the HAR equation and compared the two models using recursive forecasts from 2006 to 2025. HAR-X did not improve point forecast accuracy. Skew was nevertheless related to the size of the error made over the following 22 trading days. The distinction is important: skew appears useful for forecasting uncertainty around a HAR prediction, but not as a linear correction to the prediction itself.

1 Starting point

The data contain daily SPX returns and a 30-day option surface from January 2005 to August 2025. I measure downside skew as

$$\text{Skew}_t = \text{IV}_t^{25\Delta \text{ put}} - \text{IV}_t^{50\Delta \text{ call}}. \quad (1)$$

The option nodes have 30 calendar days to maturity. Dates without both nodes are dropped. The cleaned sample has 5,198 return observations and 5,181 option dates. Skew is expressed in decimal volatility units; its mean is 0.0353 and its median is 0.0317.

The initial forecast is a standard heterogeneous autoregressive model [1]. Daily variance is $v_t = 252r_t^2$, and the target is volatility over the following 22 trading observations:

$$RV_{t,t+22} = \sqrt{252 \left(\frac{1}{22} \sum_{j=1}^{22} r_{t+j}^2 \right)}. \quad (2)$$

While inspecting this model, I noticed that its errors became less stable during high-skew episodes. The 2008 observations are a clear example. Mean skew in 2008 was 0.0506, while the standard deviation of the HAR error was 0.194. Only 2020 had a higher annual error standard deviation, at 0.216. Figure 1 shows the same pattern using 63-day rolling estimates.

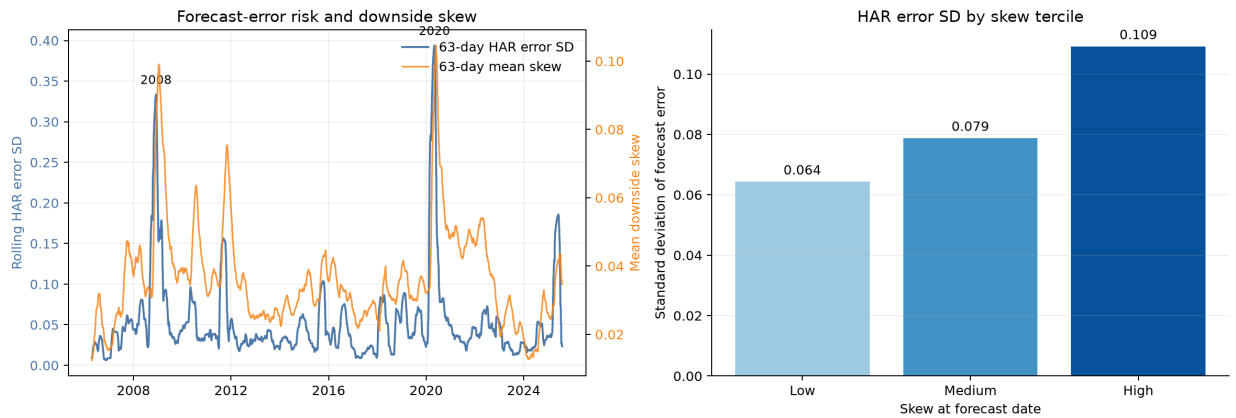


Figure 1: HAR forecast-error risk and downside skew. The left panel uses 63-observation rolling windows. The right panel groups skew observed at the forecast date into sample terciles.

The tercile calculation is less sensitive to the exact dates chosen for the crisis periods. HAR error standard deviation rises from 0.064 in the low-skew group to 0.079 in the middle group and 0.109 in the high-skew group.

2 Adding skew to HAR

The visual evidence suggested that skew might contain information missing from lagged realized variance. I added it to the forecast equation:

$$\widehat{V}_{t,t+22}^{HAR} = \beta_0 + \beta_d v_t + \beta_w \bar{v}_{t,5} + \beta_m \bar{v}_{t,22}, \quad (3)$$

$$\widehat{V}_{t,t+22}^{HAR-X} = \beta_0 + \beta_d v_t + \beta_w \bar{v}_{t,5} + \beta_m \bar{v}_{t,22} + \gamma \text{Skew}_t. \quad (4)$$

Both models are estimated with expanding-window least squares. The first forecast requires 252 valid training observations. At every date, estimation stops 22 observations before the forecast date, because later training targets would still contain unobserved returns. This produces 4,865 common forecasts from 6 March 2006 to 30 July 2025, covering most of the available twenty-year sample.

The signed forecast error is

$$e_t = \widehat{RV}_{t,t+22} - RV_{t,t+22}. \quad (5)$$

Positive bias therefore means that the model predicted too much volatility. MAE measures a typical absolute miss, while RMSE gives more weight to large misses.

3 Pseudo-out-of-sample comparison

Table 1 shows that HAR-X does not improve the point forecasts. Its bias is slightly lower, but its MAE is 1.0% higher and its RMSE is 2.7% higher. A Newey–West comparison with 21 lags gives $p = 0.225$ for the difference in absolute loss. The squared-loss difference favours HAR and has $p = 0.032$. These calculations account for serial dependence caused by overlapping 22-day targets, although the nested-model setting warrants caution when reading the p -values.

Table 1: Pseudo-out-of-sample forecast errors, March 2006–July 2025

Model	Observations	Bias	MAE	RMSE
HAR	4,865	0.01745	0.05753	0.08796
HAR-X	4,865	0.01675	0.05808	0.09032

The result is similar within the skew groups (Figure 2). HAR-X is not consistently more accurate in any part of the skew distribution. In high-skew observations, its MAE is 0.0731 compared with 0.0715 for HAR, and its RMSE is 0.1147 compared with 0.1110.

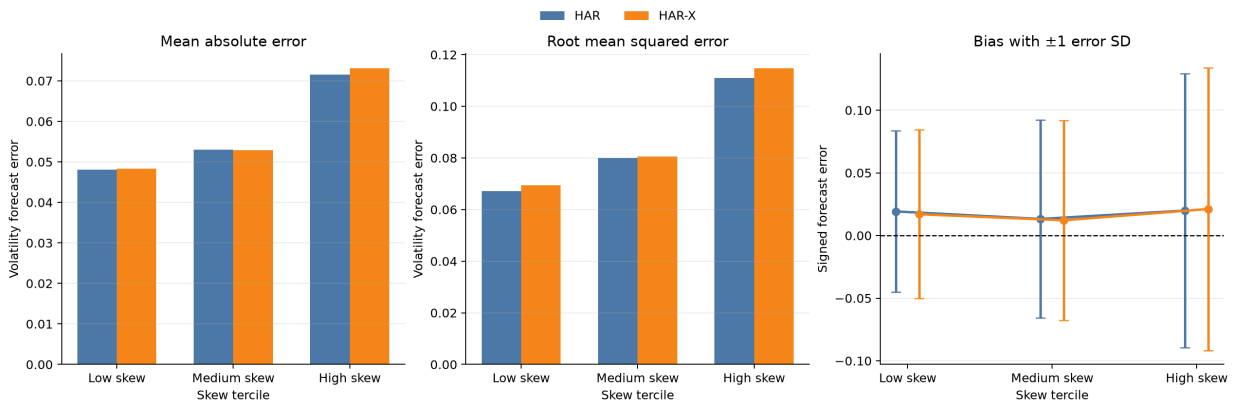


Figure 2: HAR and HAR-X forecast accuracy by skew tercile. The error bars in the third panel are one error standard deviation, not confidence intervals.

4 What skew predicts

Adding skew to the conditional mean did not help, but skew still contains information about the later error distribution. To check the tercile result more directly, I regressed the absolute HAR error and the squared HAR error on standardized skew. Newey–West standard errors use 21 lags.

A one-standard-deviation increase in skew is associated with a 0.0197 increase in absolute error (standard error 0.0046, $p < 0.001$). For squared error, the coefficient is 0.00685 (standard error 0.00202, $p < 0.001$). Since skew is observed at date t and the error depends on returns through $t + 22$, this is a predictive relationship. It is more precise to say that skew predicts error magnitude or conditional error variance than to say that it predicts an individual “error standard deviation.”

This interpretation also explains why HAR-X can fail as a point forecast. A variable can identify dates on which a forecast is unreliable without telling the model whether the subsequent realization will be above or below its current prediction. In this sample, skew appears to do the former.

5 Limits and conclusion

The analysis uses two option-surface nodes and a linear skew term. Other measures could use the full surface, changes in skew, or interactions with current volatility [3]. There is also a horizon mismatch between 30-calendar-day options and the next 22 trading returns. Finally, this is a forecast exercise rather than an option-return backtest. It does not include option prices, bid–ask spreads, hedging costs, or the distinction between a physical variance forecast and a model-free risk-neutral variance expectation [2].

The original observation is supported by the data: high skew precedes larger and more variable HAR errors, especially in 2008 and 2020. The HAR-X experiment gives a narrower result. Putting skew directly into the conditional-mean equation does not improve pointwise accuracy. Skew may instead be useful for a second-stage forecast of error variance or for sizing confidence bands around the HAR forecast.

Reproduction

```
PYTHONPATH=src .venv/bin/python src/HAR_X_comparison.py
```

The command uses `min_training=252`, prints the forecast comparison and conditional-error regressions, and creates both figures in `output/`.

References

- [1] F. Corsi, “A Simple Approximate Long-Memory Model of Realized Volatility,” *Journal of Financial Econometrics*, 7(2), 174–196, 2009. <https://doi.org/10.1093/jjfinec/nbp001>
- [2] T. Bollerslev, G. Tauchen, and H. Zhou, “Expected Stock Returns and Variance Risk Premia,” *Review of Financial Studies*, 22(11), 4463–4492, 2009. <https://doi.org/10.1093/rfs/hhn055>
- [3] G. Bakshi, N. Kapadia, and D. Madan, “Stock Return Characteristics, Skew Laws, and the Differential Pricing of Individual Equity Options,” *Review of Financial Studies*, 16(1), 101–143, 2003. <https://doi.org/10.1093/rfs/16.1.101>